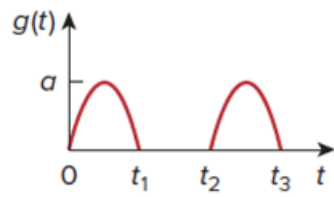


Problem

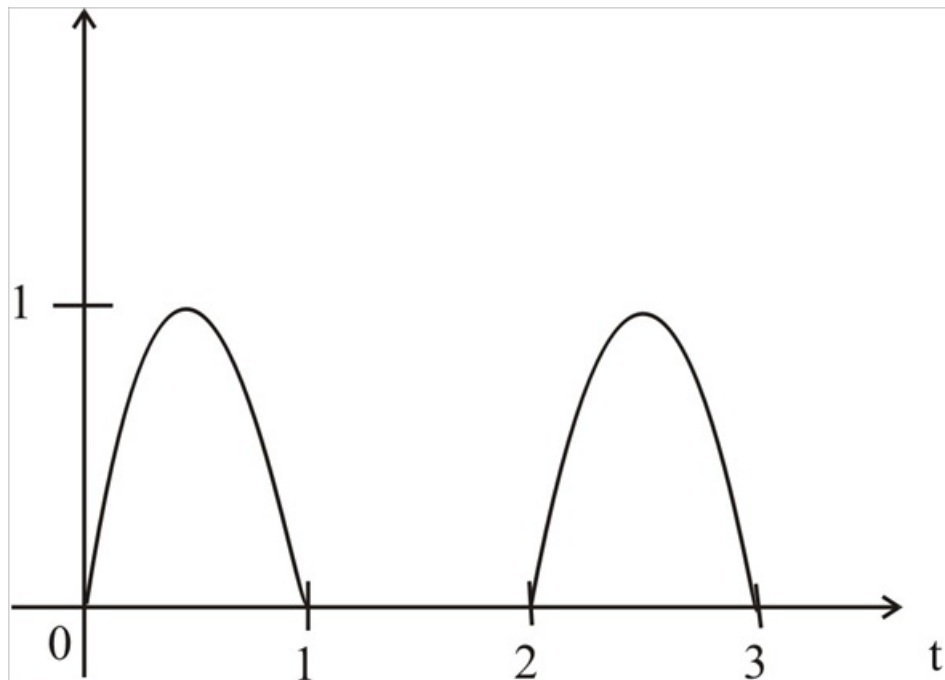
Using Fig. 15.32, design a problem to help other students better understand the Laplace transform of a simple, periodic waveshape.

Figure 15.32:



Step-by-step solution

Step 1 of 2



Step 2 of 2

Here periodic function = $g(t)$

$$\text{periodicity} = 2$$

$$\begin{aligned} G_1(B) &= \int_0^\infty g_1(t) \cdot e^{-st} dt \\ &= \int_0^\infty \sin \pi t \cdot e^{-st} dt. \\ &= \frac{\sin \pi t \cdot e^{-st}}{-s} \Big|_0^\infty + \frac{\pi}{s} \int_0^\infty \cos \pi t \cdot e^{-st} dt \\ &= \frac{-1}{s} (\sin \pi e^{-s}) + \frac{\pi}{s} \left[\frac{\cos \pi t \cdot e^{-st}}{-s} \Big|_0^\infty - \frac{\pi}{s} \int_0^\infty \sin \pi t \cdot e^{-st} dt \right] \\ &= \frac{-1}{s} (\sin \pi e^{-s}) + \frac{\pi}{s} \left[-\frac{1}{s} (\cos \pi e^{-s} - e^0) - \frac{\pi}{s} G_1(S) \right] \end{aligned}$$

$$\begin{aligned} G_1(S) \left[1 + \frac{\pi^2}{S^2} \right] &= \frac{-1}{s} \left[\sin \pi e^{-s} + \frac{\pi}{s} (\cos \pi e^{-s} - 1) \right] \\ \Rightarrow G_1(S) &= - \left[\sin \pi e^{-s} + \frac{\pi}{s} (\cos \pi e^{-s} - 1) \right] \times \frac{s}{s^2 + \pi^2} \end{aligned}$$

$$\begin{aligned} G(S) &= \frac{G^1(S)}{1 - e^{-TS}} \\ &= \frac{-s}{(s^2 + \pi^2)(1 - e^{-2s})} \left\{ e^{-s} \left(\sin \pi + \frac{\pi}{s} \cos \right) - \frac{\pi}{s} \right\} \end{aligned}$$